

Dr. Bhagyabati Moharana

Applied Scientist | PhD in Computer Science |
Machine Learning | Multimodal Analytics |
Time-Series Forecasting | SQL | Python
<https://bhagyamoharana.github.io/Portfolio/>

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PhD Applied Scientist with 6+ years of experience in AI/ML, data science, and immersive VR/XR, specialising in multimodal analytics and affective computing. Developed and validated mixed-reality systems for human performance modelling, integrating Unity-based tools to capture implicit physiological and behavioral signals. Led AI-driven multimodal analytics research using Python and R pipelines to model how physiological synchrony during collaboration predicts team performance, improving prediction accuracy by 20%. Recipient of a 48-month Industry-Co-Funding Award and international recognition for research excellence. Authored five IEEE conference papers on VR-based cognitive workload and user-state modelling, collectively cited over 20 times.

Core Skills

AI/ML: Python, R, SQL, PySpark, Scikit-Learn, TensorFlow, PyTorch, Predictive Modelling, Time-Series Forecasting, Anomaly Detection, Object Detection, Pattern Analysis, Statistical Modelling

Web-Relevant Expertise: JavaScript, CSS, HTML, Web-based Analytics Frameworks, Cloud Deployment (AWS/Azure), SaaS Integration, Scalable AI Infrastructure

Data Science and Engineering: SQL (query optimisation, wrangling, analytics), ETL Pipelines, Statistical Modelling, Feature Engineering, Data Cleaning & Preprocessing, Visualisation (Tableau, Power BI, Matplotlib, Seaborn, Plotly)

Collaboration: Industry-Academic Partnerships, Mentoring, Cross-Functional Teamwork, Speaker, Stakeholder Engagement

VR/XR Development: Unity (C#), XR Interaction Toolkit, Photon/Netcode, Digital Twins, Mixed Reality, Immersive Analytics

Technical & Scientific Writing Peer-Reviewed Publications, Research Documentation, Grant Proposals, Industry Collaborations

Work Experience

Postdoctoral Researcher– AI & Mixed Reality Systems | TUS | Ireland

2025-Present

- Led research on goal-oriented, multimodal AI systems integrating voice, text, and physiological data, achieving a 15 % gain in emotional and cognitive state prediction accuracy — aligning with Gen3’s adaptive conversational goals.
- Designed and implemented Transformer-based neural architectures and NLP models (BERT, GPT, AssemblyAI) for sentiment and engagement analysis, cutting annotation time by 35 % and improving dialogue understanding for conversational agents.
- Leveraged LLMs and prompt-engineering strategies for automated insight generation and context-aware response modelling across XR datasets.
- Deployed agentic AI assistants (Copilot, ChatGPT in Visual Studio Code) to accelerate model experimentation, code synthesis, and documentation, improving development efficiency by 40 %.
- Applied AI agent context-management to maintain adaptive memory and dynamic reasoning across iterative ML workflows — foundational to multi-agent conversational systems.
- Collaborated with 4 universities and 12 industry partners, processing 2 TB+ of multimodal data for real-time physiological analytics in human–AI interaction environments.
- Published in IEEE/ACM venues (20 + citations) advancing research at the intersection of AI, HCI, and immersive computing

Doctoral Researcher – Applied Experiments & Advanced Analytics| TUS |Ireland

2022 -2025

- Did human experiments with 68 participants in collaborative VR, expanding study complexity and duration to analyse physiological and behavioral synchrony.
- Engineered Unity + Photon systems with SQL-integrated pipelines for real-time object tracking and anomaly detection.

- Processed 10K+ object-level interactions across 1K+ collaboration events, achieving 88% precision in anomaly detection.
- Applied NLP pipelines to 50K+ conversational utterances, predicting user behavior outcomes with 78% accuracy.
- Designed real-time analytics dashboards using React, JSP, and CSS to build processing 5TB+ of multimodal data, improving accessibility and visualisation efficiency by 40%.

PhD Scholar – Foundational Research & VR Development

2019-2022

- Engineered two interactive VR environments utilising Unity and HTC Vive Pro Eye, streamlining the data collection process and generating a unique dataset of 100,000+ data points.
- Created machine learning models based on physiological synchrony to flag collaboration anomalies, utilising 30 unique data features, and achieving >85% accuracy across the entire organisation.

Early Career Experience

2011-2015

- Started an LDAP security Java portlet application at Infosys, improving enterprise authentication workflows and reducing unauthorised access attempts by 20%.
- As a Service Engineer, managed EPBX secure communication systems for 200+ enterprise users, ensuring high availability and data protection.

Research Projects

Anomaly & Pattern Detection in Multimodal Data

- Applied SQL preprocessing + ML models (DTW, r-squared) to detect irregular synchrony patterns → analysed 500,000+ physiological data points across 60 users, achieving >80% precision, 85% recall, and reducing false positives by 12%; transferable to fraud/identity attack detection.

Voice Sentiment & Behavioural Forecasting

- Built SQL-driven NLP pipeline (BERT, VADER, AssemblyAI) to process 50,000+ conversational utterances → predicted engagement outcomes with 78% accuracy and flagged negative sentiment patterns linked to 15% longer task completion times, demonstrating methods transferable to social engineering and phishing detection scenarios.

Research Publications

- "Physiological Synchrony in a Collaborative Virtual Reality Task" – IEEE Conference on VR, 2023.
- "Role-Specific Physiological Responses and Quality of Experience in Collaborative Virtual Reality Tasks" – IEEE Conference on VR, 2024, London.
- "Voices Unveiled: Quality of Experience in Collaborative VR via AssemblyAI and NASA-TLX Analysis" – IEEE Conference, 2024, Sweden.
- [Full list of publications on Google Scholar](#)

Education

Doctor of Philosophy (PhD) in Computer Science |TUS | Ireland | 2019-2025

Technical Tools and Platforms

Core Tools	:	SQL, PySpark, Python, R, Scikit-Learn, TensorFlow, PyTorch, MongoDB, Cloud Deployment (GCP/Azure), Tableau, Power BI, Github
Research Tools (additional)	:	Unity, XR Interaction Toolkit, Photon/Netcode, Digital Twins, Eye Tracking (Tobii), VR Hardware (HTC Vive, Oculus Go), Immersive Analytics

Extracurricular & Industry Engagement

- Research Collaborations with academic and industry partners for AI-driven VR collaboration and multimodal behavior analysis.
- Public Engagement in VR Research, leading demonstrations at conferences and workshops.
- Currently pursuing AWS Certified Solutions Architect – Associate (SAA-C03) to strengthen cloud deployment expertise for scalable AI analytics and SaaS product integration.